$$\hat{\rho}_{\mathcal{S}}(t) = \frac{1}{2}\left(\hat{I} + \boldsymbol{p}(t)\cdot\hat{\boldsymbol{\sigma}}\right) = \frac{1}{2}\left(\hat{I} + p_x(t)\hat{\sigma}_x + p_y(t)\hat{\sigma}_y + p_z(t)\hat{\sigma}_z\right). \quad (5.160)$$